

COMP111077 Applied Cryptography

Coursework (Semester 2: 2024 - 2025)

Component 1 – Practical Coursework (Weighting 60%)

Due Date 14th April 2025 23.59 hours

Overview

Component 1 requires you to implement solutions to given problems using either cryptographic or cryptanalytic techniques. There are 2 tasks for you complete each which assesses a different cryptographic or cryptanalysis technique. Marks indicated are indicative.

Notes for Submission

Each exercise contains multiple submissions, you must follow the file naming convention specified in the Submission section for each task. When you have completed all parts, zip them together in a file called <student_number>.zip and submit a single file containing all the exercises. Eg B001122345.zip

Task 1 – A Substitution Cipher

In the Assignments folder on Aula under Assignment Resources you each have a file with a block of ciphertext to work with. This is the format <student_id>.txt. **You should only work with your own file.** If you do not have a file with your student id, you must let the module coordinator know as soon as possible.

Your task is to decrypt the ciphertext and generate the key using cryptanalysis techniques. You may use a combination of Cryptool2 and or Python to perform this task and should clearly document the process you followed. **NB No marks will be awarded for this exercise where work cannot be verified or uses other techniques including AI based techniques such as Copilot or ChatGPT.**

Submission

You should submit the following

- A text file, <student_id>_task1.txt containing the decrypted plaintext (10 marks)
- The key, <student_id>_key.txt, this is the alphabet generated to encrypt/decrypt the cipher. This should be a direct mapping with the plaintext character first and the substitution second eg:
a -> i
b -> x
c -> j
...
z -> k
The mappings used are not case sensitive eg 'A' and 'a' would map to 'I' and 'i' in the example above. Your solution should preserve the existing case when you decrypt it. (15 marks)
- A word-processed report, <student_id>_task1.docx detailing the process you followed to

break the cipher. This should include any supporting screenshots or code clearly demonstrating the work you carried out.

(20 marks)
(Total 45 marks)

Notes

- The original plaintext uses the English language, and an alphabet limited to the ascii characters
- The cipher used is a Classic substitution cipher with a one-to-one mapping.
- Your report should include the following

A brief introduction to the problem (2 marks)

An outline of the approaches that can be taken to break the cipher (5 marks)

A detailed explanation of the approach you have taken (13 marks)

3rd party sources must be referenced using the Harvard referencing system.

Resources

Ciphertext eg B001122345.txt sent as an email attachment.

Task 2 - Simplified 7bit PDU Encoding

Using Python, write a 7 bit PDU encoder that accepts a plaintext ascii string and returns a tuple (x,y) where x is the encoded plaintext formatted as a binary sting with each byte separated with 5 hashes '#####', and y is the hexadecimal representation of the string. In devising your solution, you should not use any cryptographic libraries such as Pycryptodome or Cryptography. You can use standard Python libraries eg string.

```
Code def pdu_encode(m):
    #Your code here
    return (x,y)
    #where x is the encrypted binary representation and y is the
    encrypted hexadecimal
```

Sample Usage and Output

Usage

```
(str,lst) = pdu_encode('Welcome to applied cryptography!') print(str)
print(lst)
```

Program Output

```
11010111#####00110010#####01111011#####11111100#####01101110#####10010111#
#####01000001#####11110100#####00110111#####00101000#####00001100#####10000
111#####10110011#####11010011#####01100101#####00110010#####01101000#####0
0101100#####11001111#####11000011#####11101001#####11101111#####10110011#
###00111100#####00001100#####01000111#####11100111#####01000011#####
['\xd7', '\x32', '\x7b', '\xfc', '\x6e', '\x97', '\x41', '\xf4', '\x37',
'\x28', '\xc', '\x87', '\xb3', '\xd3', '\x65', '\x32', '\x68', '\x2c',
'\xcf', '\xc3', '\xe9', '\xef', '\xb3', '\x3c', '\xc', '\x47', '\xe7',
'\x43']
```

Submission

You should submit the following

task_two.py – A file containing your python functions. You should submit all code necessary to generate the output including any supporting functions. Credit will be given for well-structured code. If I cannot run your code, it will receive no marks.

(10 marks)

Your function will be tested against the following inputs:

- 'Hello'

(5 marks)

- 'Hello World!'

(10 marks)

- 'Welcome to applied cryptography!'

(15 marks)

(Total 40 marks)

Marking Scheme

Marks for each task are indicated at question level, the following mark scheme indicates the competencies required to assess attainment against the mark scheme.

Grade	Description
A1	Student work is exemplary and displays exceptional knowledge, understanding and analysis including evidence of a high degree of almost complete autonomy and independent judgement relative to threshold expectations
A2	Submitted work significantly exceeds the threshold for a pass at levels 11 – 12. Work displays a consistently thorough, deep and extensive level of understanding and a very high level of analysis of cryptographic principles
A3	Submitted work is very much above the threshold for a pass at levels 11 – 12. Work displays a consistently thorough level of understanding and a high level of analysis of cryptographic principles
B1	Submitted work is above the threshold for a pass at levels 11 – 12. Work displays a consistently good level of understanding and analysis of cryptographic principles
B2	Submitted work meets the threshold for a pass at levels 11 – 12. Work displays a good knowledge and understanding of cryptographic principles and good analytical and problem-solving abilities
C	Submitted work shows a satisfactory understanding and analysis of cryptographic principles.
D	Submitted work shows a partial understanding of cryptographic principles but this is incomplete. Analysis is limited.
E	Submitted work shows a limited understanding of cryptographic principles and lacks analysis
N	No submission

